

# EGRD: ASSESSING RESEARCH IDEA NOVELTY THROUGH EVIDENCE-GROUNDED RESIDUAL DECOMPOSITION

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## ABSTRACT

Language models now propose research ideas faster than anyone can vet them, which makes assessing novelty a practical bottleneck. Judging novelty requires identifying what prior work already covers and evaluating the contribution that remains. Overall similarity provides an incomplete account: an idea’s components may be covered across several papers, while their combination may still introduce a substantive change. We propose Evidence-Grounded Residual Decomposition (EGRD), which assesses novelty relative to an explicit set of temporally admissible prior works, making the basis of each judgment inspectable and allowing the effect of changing the evidence to be examined. EGRD estimates how much of the target those priors jointly cover, decomposes the target into latent semantic units, and learns a support matrix aligning these units with the evidence. It models two complementary residuals: unit residuals capture content with limited historical support, while interaction residuals capture unit combinations that no single prior supports. These residuals, together with coverage and contextual representations, predict an ordered novelty grade. A conservative blockwise aggregation rule lets a model trained with a small evidence budget read larger evidence sets at inference time without retraining. We further construct IDEADELTA, a dataset spanning graph machine learning, computer vision, and time series, pairing each idea with temporally admissible priors annotated for coverage, residual contribution, and novelty. EGRD recovers the annotation scheme consistently across the three domains, and ablations show that its predictions depend on the supplied evidence and on the residual representations rather than on the target idea alone. Transfer to expert judgments remains sensitive to where the grade boundaries are placed.

## 1 INTRODUCTION

Judging whether a research idea is new is difficult, and the growing volume of published work makes it harder. Novelty is not a monotone function of textual similarity to any single prior work, and separating a genuine advance from a restatement is in practice a judgment that draws on familiarity with the field. We therefore study *evidence-conditioned* novelty assessment: grading a target idea relative to an explicit set of prior works published before it. Fixing this set makes the basis of each judgment inspectable and turns the evidence into a variable that can be controlled.

Much existing work takes a paper as the unit of assessment, from citation-pattern measures (Uzzi et al., 2013) to document-level embeddings (Cohan et al., 2020; Ostendorff et al., 2022) and to systems that review a submission as a whole (Zhu et al., 2025). A paper bundles contributions of uneven novelty: a central mechanism may sit beside incidental variants, and one score over the bundle does not say which contribution it is about. The decision of what to pursue is made one idea at a time.

Systems that do operate on ideas assemble the judgment as a language-model workflow, retrieving related work and then prompting a model to compare against it (Wang et al., 2024; da Silva et al., 2025; Afzal et al., 2026). Both halves are fragile. The evidence is selected by the system being evaluated, so a verdict cannot be attributed to the retrieval or to the comparison, and neither can be

measured or improved on its own. The comparison itself is elicited by a prompt rather than learned against supervision. Schopf & Färber (2026) evaluate this arrangement against expert judgments and report that although the reasoning such models produce closely mirrors human rationales, that alignment does not reliably translate into accurate novelty judgments, which diverge significantly from the expert gold standard.

Such a workflow also compresses the comparison into a single score, and three distinctions do not survive that compression. Historical coverage can be *dispersed*: several papers may jointly account for an idea even when no single paper closely resembles it, so a comparison against the nearest prior work misses it. Ideas at equal coverage differ in *what remains*: one may leave only an implementation detail unexplained while another introduces a new mechanism. And components that are individually covered may form a *combination* that no single prior work puts together. Each is a question about a particular part of the idea and the particular prior work responsible for it, and a score that discards this correspondence cannot be post-processed to recover it. Assessing novelty calls instead for a representation that keeps it: which prior work accounts for which part of the target, and what is left over.

Studying this setting requires ideas paired with the evidence admissible at the time they were proposed, which existing novelty benchmarks do not provide. We therefore construct IDEADELTA, a dataset spanning graph machine learning, computer vision, and time series, in which every entry is an idea-length statement paired with temporally admissible priors and annotated at both the idea and target-prior levels. The pairing supports the proposed task and makes controlled changes to the supplied evidence possible.

We propose Evidence-Grounded Residual Decomposition (EGRD), which decomposes a target against ranked historical evidence into what that evidence accounts for and what it leaves, and predicts an ordered novelty grade from the two. For larger evidence sets, a conservative blockwise inference rule extends the evidence budget without retraining. Our contributions are:

- **IDEADELTA, a multi-domain, evidence-paired dataset.** We provide ideas paired with temporally admissible priors and annotations of coverage, residual contribution, and novelty across three research areas. Because the evidence is supplied rather than retrieved, it can be intervened on: permuting the priors while holding the target fixed drops the model to near-chance in every domain.
- **Unit-level coverage over dispersed evidence.** We model how much of the target the supplied priors account for at the level of latent semantic units rather than of the idea as a whole. A learned support matrix aligns each unit with each prior, and noisy-OR aggregation lets several partially relevant papers jointly cover a unit that none of them covers alone.
- **Typed residuals.** We represent what the priors leave unaccounted for as two distinct quantities: unit residuals, for content that no prior supports, and interaction residuals, for combinations of individually covered components that no single prior puts together. Both are read off the same support matrix, so the distinction between an unexplored contribution and an effective recombination is produced by the model rather than asserted.

## 2 RELATED WORK

**Automated research and idea generation.** Language models are increasingly used to carry out parts of the research process itself. Wang et al. (2024) retrieve inspirations from past literature and optimise explicitly for novelty by comparing a candidate against prior papers; Baek et al. (2025) refine proposals against a set of LLM reviewing agents that give structured feedback along several dimensions; and Lu et al. (2024) close the loop from ideation through experiments to a written paper and a simulated review. Si et al. (2024) evaluate what such pipelines produce, finding the generated ideas rated more novel than human-written ones but less feasible. Every one of these systems contains an internal novelty check, and it is the check rather than the generator that bounds them: once proposals are cheap, the output of the pipeline is whatever its filter admits. The filter is nonetheless the component these systems evaluate least directly, judged only through the ideas that eventually emerge downstream of it.

**Judging the novelty of scientific work.** Studied on its own, the check has passed through several generations of method. The earliest operate on citation statistics, measuring how unusual the combinations of references in a paper are against historical co-occurrence (Uzzi et al., 2013); they require the work to already exist and be cited, and reduce novelty to patterns of co-occurrence rather than to what a paper says. Document-level semantic representations replace surface statistics with learned relatedness, scoring how close a submission is to prior papers through citation-informed embeddings (Cohan et al., 2020; Ostendorff et al., 2022), but they compare documents as points and recover nothing about which content is shared. Most recent systems retrieve related literature and hand the comparison to a language model. Liang et al. (2024) obtain structured feedback from a single prompt, of which significance and novelty is one section, and report partial overlap with human reviews; Zhu et al. (2025) train a reviewer model on a multi-stage process of structured analysis, retrieval, and evidence-based argumentation; Afzal et al. (2026) split the judgment into claim extraction, retrieval and synthesis of related work, and structured comparison, motivating the design with an analysis of human-written novelty reviews; and da Silva et al. (2025) build an interactive system on the same retrieve-then-compare pattern. Schopf & Färber (2026) supply the evaluation counterpart and report that model ratings still diverge substantially from expert ones even when the accompanying reasoning appears sound. Two properties persist across these generations, and both are structural rather than incidental. The evidence is retrieved by the system under evaluation, so its verdicts are unattributable: an error cannot be assigned to the retrieval or to the judgment, and the two therefore cannot be improved, or even measured, separately. We remove this by conditioning on a given, temporally admissible evidence set, which additionally turns the evidence into a variable that can be intervened on. The second property is that the target is scored as an undivided whole, against the closest prior work or against a retrieved set read as a block. Such a comparison returns one number per target, and no post-processing of that number can recover which part of the idea it is about; the information is discarded before the score is formed, not lost afterwards.

**Decomposed verification against evidence.** Saying which part remains requires decomposing the target, and outside novelty assessment this has become the standard recipe for factuality evaluation: break a text into smaller propositions and check each against retrieved evidence. Min et al. (2023) decompose a passage into atomic facts, verify each against a knowledge source, and report the supported fraction; Wei et al. (2024) scale the same procedure with search as the evidence source; and in the scientific domain the corresponding task checks a stated claim against candidate abstracts while identifying the supporting rationales (Wadden et al., 2020; 2022). Three properties of the recipe do not survive the move to research ideas. The propositions are assumed to be given, whereas an idea carries no stated claims, and the decomposition step is not a neutral preprocessor: Wanner et al. (2024) show that scores of this family shift materially with the decomposition used. Support is treated as a label, whereas a prior work can cover part of a unit and not the rest. And each unit is verified independently before the results are averaged, so the score is a function of the per-unit coverages alone and cannot express whether a combination of units has precedent. We accordingly induce the units rather than reading them off the text, measure support as a graded quantity, and add a residual defined over pairs of units.

### 3 DATASET

Novelty can only be assessed relative to earlier work. Each evaluation example must therefore specify a target idea, the prior ideas available when it was proposed, and the contribution that remains after accounting for those priors. We construct IDEADELTA to provide these three elements.

#### 3.1 TASK AND LABELS

The unit of annotation is a research *idea*: one innovation claim extracted from a paper. For example, a paper that introduces both a new objective and a new architecture contributes two ideas, which are evaluated separately. Let  $x$  denote a target idea published at time  $t_x$ . Its admissible priors are  $\mathcal{P}_x = \{p_k \mid t_{p_k} < t_x\}$ . This temporal constraint is essential: an idea published after the target cannot serve as evidence about the target’s novelty, regardless of its semantic similarity. We annotate each target–prior pair with one of four coverage levels and each target with an ordered novelty grade from LOW to HIGH.

Table 1: Dataset composition. Counts of ideas and of target–prior pairs, and the idea-level novelty distribution. Pairs are formed within a source domain only.

|          | Graph ML | Computer vision | Time series | All    |
|----------|----------|-----------------|-------------|--------|
| Ideas    | 5,225    | 8,291           | 7,595       | 21,111 |
| Pairs    | 21,183   | 36,238          | 35,455      | 92,876 |
| LOW      | 819      | 901             | 1,724       | 3,444  |
| WEAK     | 1,240    | 1,711           | 1,614       | 4,565  |
| MODERATE | 2,247    | 2,777           | 2,173       | 7,197  |
| HIGH     | 919      | 2,902           | 2,084       | 5,905  |

Rather than assign novelty as a single overall impression, we derive it from four annotated quantities:

$$\text{raw}(x) = (1 - C_x) \cdot \Delta_x \cdot E_x \cdot (1 - \lambda S_x), \quad \lambda = 0.45, \quad (1)$$

where  $C_x$  measures how much of the target is jointly covered by the retained priors,  $\Delta_x$  measures the substantive contribution that remains,  $E_x$  measures whether the retrieved evidence is sufficient for assessment, and  $S_x$  measures how much of the claimed difference is merely surface-level. We convert the resulting score into the four ordered novelty grades. The rule assigns lower novelty when prior work already covers the target, when the remaining difference is not substantive, when the available evidence is insufficient, or when the difference consists mainly of renaming or other surface changes.

### 3.2 CONSTRUCTION

**Paper selection and idea extraction.** We collect arXiv papers from graph machine learning, computer vision, and time series. We prioritize papers that were subsequently published, especially at major venues. An LLM extracts innovation claims from the title, abstract, and introduction. A second pass removes statements that do not describe a substantive research contribution, including open-source announcements, bare state-of-the-art claims, and engineering deliverables.

**Prior retrieval.** For each target, nearest-neighbor search over idea embeddings retrieves candidate priors published before the target. An LLM adjudicates candidates in the uncertain similarity range. We also perform a cross-type retrieval pass so that a target can be linked to relevant prior work of another contribution type. All target–prior pairs remain within the same research domain.

**Annotation.** The same judge, `deepseek-v4-pro`, annotates all stages. It first assigns pairwise coverage labels, with ambiguous cases judged from their technical content. It then estimates joint coverage, the remaining substantive contribution, evidence sufficiency, and surface-level difference for each target. Equation (1) converts these quantities into the idea-level novelty grade. The labels are model-generated, and we manually inspect samples from each stage. Appendix A.1 describes the full pipeline.

### 3.3 COMPOSITION

IDEADELTA contains 21,111 annotated ideas and 92,876 annotated target–prior pairs, with target dates ranging from 2019-01 to 2026-09 (Table 1). Each idea records its source domain, and the *All* column is the union of the three domain subsets rather than an additional collection. We order the train, validation, and test splits by date, with cut points at 2025-02 and 2025-11. The resulting splits contain 59,269/16,436/17,171 pairs and 13,815/3,944/3,352 ideas.

Because the labels are model-generated, we further test whether they behave as novelty labels should. Appendix A.2 examines whether metadata can predict the labels, whether the retrieved prior content matters, and whether the assigned novelty changes appropriately when the evidence is modified.

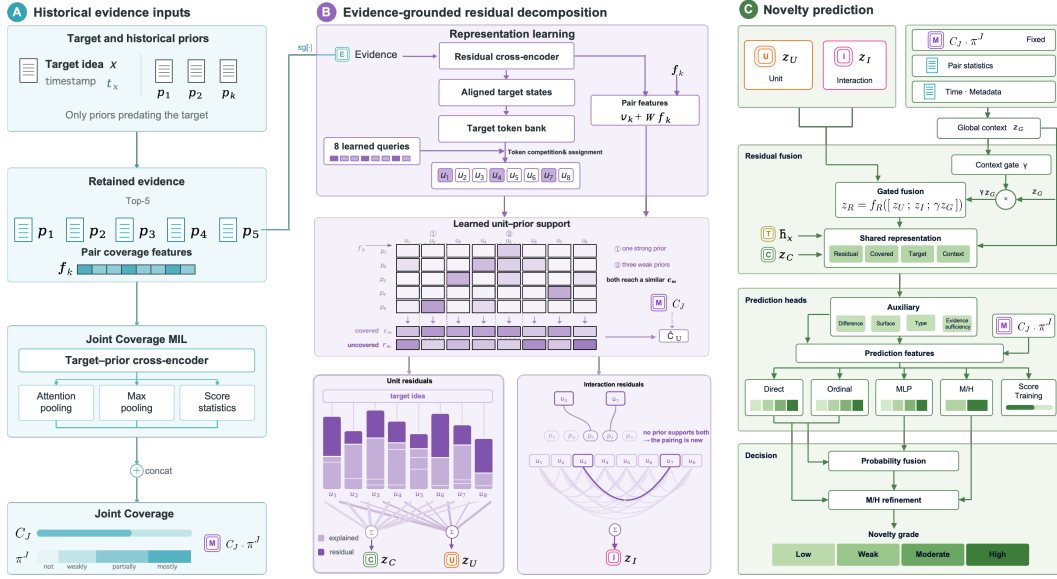


Figure 1: Overview of EGRD. (A) Only priors predating the target are admissible; the retained top- $K$  evidence carries pair coverage features  $f_k$ , and a multiple-instance model over target-prior pairs predicts joint coverage  $C_J$  and its four-class distribution  $\pi^J$ . These outputs enter the later stages as fixed inputs, marked sg[·]. (B) A residual cross-encoder builds a target token bank, which  $M$  learned queries partition into semantic units  $u_m$ . A learned support matrix  $A_{km}$  aligns each unit with each prior; noisy-OR over priors gives unit coverage  $c_m$ , and the complementary residual  $r_m$  weights the units into  $z_U$ , with the covered part pooled into  $z_C$ . The support matrix also yields co-support  $b_{ij}$ , so a unit pair that no single prior supports jointly contributes to the interaction residual  $z_I$ . (C) A gate modulates global context within residual fusion; the fused representation, the covered and target representations, and global context form a shared representation. Auxiliary predictions and joint coverage further inform the novelty branches. Direct and ordinal probabilities enter both the fused distribution and the moderate-to-high refinement, while the continuous score head supplies auxiliary training supervision and is excluded from the inference-time grading rule.

## 4 METHODOLOGY

Given a target research idea and a set of prior ideas proposed before it, EGRD assesses the target’s novelty relative to the supplied evidence. The model proceeds in three stages, shown in Figure 1. Stage A assembles the historical evidence: it selects temporally admissible priors and summarizes how much of the target they already account for. Stage B decomposes the target against that evidence and represents the uncovered contribution through two complementary components, where unit residuals capture semantic content with insufficient historical support and interaction residuals capture potentially novel combinations of semantic units. Stage C predicts an ordered novelty grade from LOW to HIGH from these residual representations.

Throughout,  $[\cdot; \cdot]$  denotes concatenation,  $\odot$  elementwise multiplication,  $d$  the encoder dimension,  $\sigma$  the sigmoid function, and  $\varepsilon$  a small positive constant added for numerical stability. Padded priors are masked out, and  $\mathcal{V}$  denotes the non-padding positions of the target.

### 4.1 STAGE A: HISTORICAL EVIDENCE INPUTS

Stage A assembles historical evidence and estimates how much of the target idea that evidence collectively covers. Given a target idea  $x$  with timestamp  $t_x$ , we restrict candidate priors to  $\mathcal{P}_x = \{p_k \mid t_{p_k} < t_x\}$ , where  $t_{p_k}$  denotes the timestamp of prior idea  $p_k$ . We retain the top- $K$  ranked priors, or all candidates if fewer are available, to form the evidence set  $\mathcal{E}_x \subseteq \mathcal{P}_x$  ( $K = 5$  in our experiments). Each retained prior  $p_k$  is associated with coverage features  $f_k$  for the target-prior pair.

To estimate coverage across the evidence set, we use a multiple-instance model (Ilse et al., 2018). A cross-encoder encodes each target–prior pair, and the model aggregates the pair representations through attention pooling and maximum pooling. The pooled representations are concatenated with evidence-score statistics to predict joint coverage  $C_J$  and a four-class coverage distribution  $\pi^J$ . Appendix B.1 specifies the pair-scoring rule and the joint model. The retained evidence, pair features, and coverage predictions serve as fixed inputs to Stages B and C, grounding the residual decomposition and providing coverage information for novelty prediction.

We summarize the evidence set in a global context vector  $z_G$ , a learned projection of twenty-one scalars—the scaled target timestamp, pair-score statistics over  $\mathcal{E}_x$ , and the joint coverage score, entropy, and distribution  $\pi^J$ —concatenated with embeddings of two target facets (aspect and contribution type). Apart from the timestamp and the facets,  $z_G$  is a re-encoding of the Stage A outputs, and Stage C consumes it as a summary of what the evidence collectively establishes.

#### 4.2 STAGE B: EVIDENCE-GROUNDED RESIDUAL DECOMPOSITION

Stage B decomposes the target into semantic units and estimates their support from the retained evidence. It represents the unexplained contribution through two complementary components: unit residuals for content with limited historical support, and interaction residuals for combinations with limited joint support from any single prior.

**Semantic decomposition.** We use a residual cross-encoder initialized from the Stage A model and trained separately from it. For each target–prior pair, the encoder produces a pooled representation  $v_k$ , which we augment with projected pair coverage features:  $\tilde{v}_k = v_k + f_P(f_k)$ . To obtain a shared representation of the target, we average its contextual token states at matching positions across valid pairs, forming a token bank  $\{h_j\}_{j=1}^N$ , where  $N$  is the number of target tokens. We then use  $M$  learned queries  $q_m$  to softly assign these tokens to semantic units (Locatello et al., 2020) ( $M = 8$  in our experiments). Queries compete for each token through a softmax over units; the resulting assignments are then normalized over tokens within each unit:

$$B_{mj} = \text{softmax}_m \left( \frac{q_m^\top h_j}{\sqrt{d}} \right), \quad \alpha_{mj} = \frac{B_{mj}}{\sum_{j' \in \mathcal{V}} B_{mj'} + \varepsilon}, \quad \mathbf{u}_m = \sum_{j \in \mathcal{V}} \alpha_{mj} h_j. \quad (2)$$

The units have no predefined semantic roles. A learned activity score  $v_m = \sigma(\mathbf{w}_v^\top \mathbf{u}_m + b_v)$  controls each unit’s contribution, while regularization discourages redundant token assignments.

**Evidence-aligned unit residuals.** We estimate how strongly each prior  $p_k$  supports each unit  $\mathbf{u}_m$  through a learned support matrix:

$$A_{km} = \sigma \left( f_A([\mathbf{u}_m; \tilde{v}_k; \mathbf{u}_m \odot \tilde{v}_k; |\mathbf{u}_m - \tilde{v}_k|; \mathbf{f}_k]) \right), \quad (3)$$

where  $f_A$  is an MLP. Support is estimated separately for every target–prior pair, so different priors can account for different units. For  $n = |\mathcal{E}_x|$  retained priors, noisy-OR aggregates support across priors into unit coverage  $c_m$ . Its complement, weighted by unit activity, defines the residual weight  $r_m$ :

$$c_m = 1 - \prod_{k=1}^n (1 - A_{km}), \quad r_m = v_m(1 - c_m), \quad \mathbf{z}_U = \frac{1}{M} \sum_m r_m \mathbf{u}_m. \quad (4)$$

Noisy-OR lets several partially relevant priors jointly cover a unit that none of them covers alone. We use it as a differentiable aggregation rule, not as a claim that the priors are independent sources of support (Appendix B.2). An active unit with little historical support receives a large residual weight in  $\mathbf{z}_U$ . We normalize by the fixed number of units  $M$  rather than by  $\sum_m r_m$ , so that the pooled representation retains the total residual mass instead of dividing it away. We also pool the supported content as  $\mathbf{z}_C = M^{-1} \sum_m v_m c_m \mathbf{u}_m$ . To connect unit coverage to supervision, a learned, activity-weighted pooling of  $c_m$  reconstructs joint coverage  $\hat{C}_U$ , supervised by annotated coverage and the fixed Stage A prediction. The entries of the support matrix are latent strengths learned without unit-level alignment labels.

**Interaction residuals.** Units that are individually covered may still form a new combination. For each unit pair  $i < j$ , an MLP maps their concatenation, elementwise product, and absolute difference to an interaction representation  $\mathbf{t}_{ij}$ , and a sigmoid head predicts its strength  $g_{ij}$ . We measure historical co-support by the largest product of the two units’ support strengths within a single prior. The co-support score  $b_{ij}$ , interaction residual weight  $r_{ij}$ , and pooled representation are

$$b_{ij} = \max_k A_{ki} A_{kj}, \quad r_{ij} = g_{ij}(1 - b_{ij}), \quad \mathbf{z}_I = \sum_{i < j} \eta_{ij} r_{ij} \mathbf{t}_{ij}, \quad (5)$$

where the learned attention  $\eta_{ij}$  includes a  $\log(r_{ij} + \varepsilon)$  bias that favors pairs with larger residual weights. When different priors support the two units but no single prior strongly supports both, their unit residuals can be small while their interaction residual remains large, provided that  $g_{ij}$  is high. An auxiliary classifier on  $\mathbf{z}_I$  is trained with idea-level labels for nontrivial combinations. Co-support serves as a proxy for an existing combination: a prior that supports both units need not cover the specific way they are combined in the target. Appendix B.2 provides the detailed pooling equations.

### 4.3 STAGE C: NOVELTY PREDICTION

Stage C predicts an ordered novelty grade from the residual representations and the evidence context. To account for evidence availability and uncertainty, we condition residual fusion on the global context  $\mathbf{z}_G$  defined in Section 4.1. A learned gate  $\gamma$  takes  $\mathbf{z}_U$ ,  $\mathbf{z}_I$ ,  $\mathbf{z}_G$ , and the entropy of the Stage A coverage prediction as inputs and scales the context within the residual branch:

$$\mathbf{z}_R = f_R([\mathbf{z}_U; \mathbf{z}_I; \gamma \mathbf{z}_G]), \quad \mathbf{h} = \text{LayerNorm}([\mathbf{z}_R; \mathbf{z}_G; \bar{\mathbf{h}}_x; \mathbf{z}_G]), \quad (6)$$

where  $\bar{\mathbf{h}}_x$  is obtained by masked mean pooling over the target token bank. The shared representation  $\mathbf{h}$  retains the covered content, the target representation, and a direct path for global context. The gate therefore controls context only within residual fusion.

Auxiliary heads predict substantive difference, surface change, evidence sufficiency, and residual type. These predictions, together with  $\mathbf{h}$  and joint coverage, form the inputs to three novelty branches: a direct linear classifier, a cumulative ordinal head that models grade order, and a two-layer MLP classifier. Each branch produces a four-class distribution, and their convex mixture determines an initial grade. A moderate-to-high refinement upgrades an initial MODERATE prediction when a weighted combination of the direct and ordinal HIGH-class probabilities and a separate binary-head score exceeds a threshold. The mixture weights and refinement threshold are selected on development data and fixed for testing (Appendix B.3). A separate continuous novelty head provides auxiliary supervision during training; its output is not used in the inference-time grading rule.

### 4.4 TRAINING OBJECTIVE

We initialize the residual encoder from the Stage A joint-coverage checkpoint and jointly optimize it with the decomposition and prediction modules in Stages B and C, while keeping the Stage A models fixed. The objective combines classification of the novelty grade and the residual type, supervision of a family of annotated scalar quantities, and regularization of the decomposition.

Let  $\mathcal{A} = \{\Delta, S, E, C, \nu\}$  index the annotated scalars—the substantive difference, surface change, and evidence sufficiency read off the auxiliary heads, the reconstructed coverage  $\hat{C}_U$ , and the continuous novelty score. Each contributes a smooth- $L_1$  level loss  $\mathcal{L}_a$  against its annotation. Level losses alone leave these heads free to settle on the median of their target, so we also supervise their order: the ranking terms of  $\Delta$ ,  $S$ ,  $E$ , and  $C$  are collected into a single  $\mathcal{L}_{\text{rank}}$ , while the novelty score carries its own ranking term inside  $\mathcal{L}_\nu$ . The objective is

$$\mathcal{L} = \sum_{a \in \mathcal{A}} \lambda_a \mathcal{L}_a + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}} + \lambda_{\text{nov}} \mathcal{L}_{\text{nov}} + \lambda_{\text{type}} \mathcal{L}_{\text{type}} + \lambda_{\text{dec}} \mathcal{L}_{\text{dec}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}}. \quad (7)$$

Here  $\mathcal{L}_{\text{nov}}$  supervises the three novelty branches, their fused distribution, and the moderate-to-high head, and  $\mathcal{L}_{\text{type}}$  supervises residual categories and nontrivial combinations. Beyond its level loss, the coverage term  $\mathcal{L}_C$  adds a consistency loss against the fixed Stage A prediction, assigning less consistency weight to predictions with higher entropy.  $\mathcal{L}_{\text{dec}}$  regularizes the decomposition against redundant token assignments and unused units, and  $\mathcal{L}_{\text{var}}$  is a variance floor that keeps the unit coverages from coinciding. Appendix B.4 defines each term and reports the coefficients  $\lambda$ .

Table 2: Panel A<sup>†</sup>. Baselines and our model on the IdeaDelta evaluation set, 3,177 ideas, same recalibrated labels and the same idea-only protocol (top-5 retrieved priors, no paper text), split by domain. Each prompt-based system appears once per backbone, so a row group isolates the effect of the model behind an unchanged prompt chain. Within any row the three domains move together, while the gap between rows is an order of magnitude larger. A majority-class predictor that ignores the input scores 0.312, 0.356 and 0.328 accuracy on the three domains, with quadratic weighted kappa of zero by construction; several rows fall below it.

| System                 | Backbone        | Graph ML     |              |              |              |              | Computer vision |              |              |              |              | Time series  |              |              |              |              |
|------------------------|-----------------|--------------|--------------|--------------|--------------|--------------|-----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                        |                 | Acc          | Ma-F1        | BalAcc       | QWK          | MAE ↓        | Acc             | Ma-F1        | BalAcc       | QWK          | MAE ↓        | Acc          | Ma-F1        | BalAcc       | QWK          | MAE ↓        |
| Direct, 5 priors       | Qwen3.5         | 0.300        | 0.266        | 0.336        | 0.319        | 0.925        | 0.308           | 0.264        | 0.331        | 0.300        | 0.876        | 0.291        | 0.283        | 0.352        | 0.327        | 0.956        |
|                        | deepseek-flash  | 0.438        | 0.444        | 0.476        | 0.558        | 0.706        | 0.409           | 0.418        | 0.451        | 0.521        | 0.728        | 0.416        | 0.415        | 0.440        | 0.560        | 0.716        |
|                        | gpt-5.6-luna    | 0.432        | 0.439        | 0.438        | 0.579        | 0.657        | 0.420           | 0.430        | 0.439        | 0.561        | 0.661        | 0.445        | 0.436        | 0.443        | 0.576        | 0.653        |
| RINoBench              | Qwen3.5         | 0.324        | 0.286        | 0.330        | 0.335        | 0.831        | 0.346           | 0.293        | 0.340        | 0.338        | 0.777        | 0.344        | 0.333        | 0.370        | 0.413        | 0.793        |
|                        | deepseek-flash  | 0.443        | 0.458        | 0.451        | 0.560        | 0.639        | 0.416           | 0.411        | 0.417        | 0.501        | 0.676        | 0.442        | 0.447        | 0.448        | 0.553        | 0.654        |
|                        | gpt-5.6-luna    | 0.391        | 0.377        | 0.389        | 0.487        | 0.698        | 0.393           | 0.364        | 0.381        | 0.458        | 0.683        | 0.392        | 0.379        | 0.395        | 0.512        | 0.688        |
| UKPLab 3-stage         | Qwen3.5         | 0.280        | 0.161        | 0.271        | 0.125        | 0.928        | 0.289           | 0.150        | 0.265        | 0.106        | 0.873        | 0.274        | 0.180        | 0.285        | 0.132        | 0.973        |
|                        | deepseek-flash  | 0.351        | 0.267        | 0.306        | 0.301        | 0.761        | 0.376           | 0.272        | 0.305        | 0.259        | 0.741        | 0.386        | 0.289        | 0.322        | 0.290        | 0.727        |
|                        | gpt-5.6-luna    | 0.318        | 0.226        | 0.283        | 0.266        | 0.800        | 0.367           | 0.248        | 0.304        | 0.304        | 0.721        | 0.365        | 0.267        | 0.318        | 0.296        | 0.743        |
| GraphMind <sup>§</sup> | Qwen3.5         | 0.264        | 0.205        | 0.281        | 0.154        | 1.027        | 0.264           | 0.182        | 0.253        | 0.108        | 0.959        | 0.244        | 0.204        | 0.282        | 0.155        | 1.051        |
|                        | deepseek-flash  | 0.261        | 0.217        | 0.289        | 0.154        | 1.111        | —               | —            | —            | —            | —            | —            | —            | —            | —            | —            |
|                        | gpt-5.6-luna    | 0.366        | 0.346        | 0.351        | 0.346        | 0.778        | —               | —            | —            | —            | —            | —            | —            | —            | —            | —            |
| DeepReviewer           | DeepReviewer-7B | 0.294        | 0.237        | 0.262        | 0.034        | 0.965        | 0.325           | 0.235        | 0.262        | −0.043       | 0.972        | 0.296        | 0.222        | 0.249        | −0.017       | 0.964        |
| <b>EGRD (ours)</b>     | —               | <b>0.797</b> | <b>0.798</b> | <b>0.802</b> | <b>0.908</b> | <b>0.204</b> | <b>0.763</b>    | <b>0.770</b> | <b>0.780</b> | <b>0.882</b> | <b>0.239</b> | <b>0.786</b> | <b>0.772</b> | <b>0.783</b> | <b>0.896</b> | <b>0.218</b> |

Table 3: Panel B. RINoBench test, 277 ideas, human expert 1–5 labels folded to four levels with 4/5 merged. The merged top class holds 41.5% of the 277 ideas, so predicting it everywhere scores 0.4152 accuracy, at or above every system in the table; accuracy carries no information here. Our row reaches that same 0.4152 while placing 16/36/76/149 ideas across the four classes against a gold 15/60/87/115, so the accuracy is matched by a prediction that is not constant. Read Ma-F1, BalAcc and QWK. Bold marks the best value in a column; Acc and MAE are not led by our row.

| System                                                                      | Acc           | Ma-F1         | BalAcc        | QWK           | MAE ↓         |
|-----------------------------------------------------------------------------|---------------|---------------|---------------|---------------|---------------|
| <i>Same Qwen3.5 backbone, zero-shot</i>                                     |               |               |               |               |               |
| Qwen3.5-397B-A17B                                                           | 0.3574        | 0.2244        | 0.2563        | 0.1451        | 0.7726        |
| GraphMind                                                                   | 0.3357        | 0.2281        | 0.2748        | 0.1469        | 0.7653        |
| UKPLab 3-stage                                                              | 0.3466        | 0.1892        | 0.2706        | 0.0808        | <b>0.7220</b> |
| <i>Released predictions of the benchmark authors, folded to four levels</i> |               |               |               |               |               |
| GPT-5                                                                       | 0.3646        | 0.2457        | 0.2573        | 0.0955        | 0.8051        |
| o3                                                                          | 0.3971        | 0.2370        | 0.2669        | 0.0606        | 0.8087        |
| gpt-oss-120b                                                                | 0.4152        | 0.2249        | 0.2681        | 0.0392        | 0.8375        |
| Llama-3.1-8B-Instruct                                                       | 0.3755        | 0.1961        | 0.2408        | −0.0011       | 0.8773        |
| DeepSeek-R1                                                                 | <b>0.4224</b> | 0.1873        | 0.2606        | 0.0283        | 0.8700        |
| Llama-4-Scout-17B-16E                                                       | 0.4116        | 0.1864        | 0.2548        | 0.0215        | 0.8809        |
| Llama-3.3-70B-Instruct                                                      | 0.4188        | 0.1531        | 0.2529        | 0.0051        | 0.9025        |
| <i>Reviewer model fine-tuned on paper review, local weights</i>             |               |               |               |               |               |
| DeepReviewer-7B                                                             | 0.1913        | 0.1425        | 0.2756        | 0.0145        | 1.3610        |
| <b>EGRD (ours)<sup>†</sup></b>                                              | 0.4152        | <b>0.3407</b> | <b>0.3475</b> | <b>0.2245</b> | 0.8087        |

## 5 CONSOLIDATED RESULTS

Except where a table names another benchmark, the results in this section are computed on the test split of IDEADELTA (Section 3).

**Table notes.** <sup>†</sup> Not zero-shot, and one configuration rather than a per-metric best. The row is a single run with its own confusion matrix: the substantive-delta score, pooled by taking the minimum over 60 priors read as twelve disjoint blocks of five, with three decision cuts. The weights are frozen, but the aggregation rule and the cuts were fitted on the RINoBench train split of 1,104 ideas. The

Table 4: Per-class F1 of EGRD on the three domain test splits. The ranking of the four classes is identical in every domain: LOW is easiest, MODERATE is hardest, and HIGH sits between them. The weak point is a property of the label boundary, not of any one field.

| Test split           | LOW    | WEAK   | MODERATE | HIGH   |
|----------------------|--------|--------|----------|--------|
| Graph ML             | 0.916  | 0.782  | 0.695    | 0.797  |
| Computer vision      | 0.881  | 0.778  | 0.672    | 0.748  |
| Time series          | 0.914  | 0.729  | 0.685    | 0.757  |
| All (IdeaDelta test) | 0.9045 | 0.7601 | 0.6795   | 0.7628 |

LLM rows received no such fitting, so the two blocks are not a like-for-like comparison. An oracle remapping of the LLM integer outputs moves them by at most +0.005 Macro-F1, so the gap is not calibration alone, but the protocols differ and are reported separately. The cuts are placed by a rule that must be stated in full because it carries a free parameter and that parameter matters. Earlier versions of this table matched the predicted class proportions to the gold proportions on the train split, which optimises no metric in particular. We instead maximise accuracy on that split subject to its Macro-F1 staying within a slack  $\epsilon$  of what distribution matching achieves there, and apply the resulting three values to the 277 test ideas unchanged. The row above uses  $\epsilon = 0.01$ . Sweeping  $\epsilon$  from 0 to 0.03 moves test accuracy over 0.3718–0.4224 and Macro-F1 over 0.3042–0.3407, and  $\epsilon = 0.01$  is the value at which Macro-F1 and BalAcc peak, so we did not select it independently of the test split and it should not be read as a pre-registered choice. The full sweep is in the appendix. A parameter-free variant of the same idea, maximising accuracy plus Macro-F1 on the train split with no slack, needs no such caveat and gives 0.4043 / 0.3381 / 0.3430 / 0.2257 / 0.8159 at 25 priors; readers who want a number with no tuned quantity behind it should use that one. What is robust across every variant is the direction. A paired bootstrap over 4,000 resamples of the test ideas puts the accuracy gain of the new cut rule over distribution matching at +0.054, 95% CI [+0.011, +0.098], while Macro-F1 and QWK shift by +0.022 and +0.010 with intervals that cross zero. The rule buys accuracy; it does not buy Macro-F1 or QWK, and it does not trade them away either. <sup>‡</sup> Label rule shared by Tables 2 and 4. The dataset label is a threshold on  $\text{raw} = (1 - \text{coverage}) \cdot \text{delta} \cdot \text{evidence} \cdot (1 - 0.45 \text{ surface})$ . The rule inherited from the single-domain dataset additionally gated the HIGH class on  $\text{evidence} \geq 0.72$ , and on this dataset that gate is inverted: evidence correlates with joint coverage at +0.817, so the gate admits 6,853 rows whose mean coverage is 0.724 and whose mean raw is 0.075, against 0.204 for the rows it excludes. The gate left 160 HIGH rows out of 25,435. The IdeaDelta label rule drops the gate and rescales the three cuts from 0.42/0.24/0.10 to 0.256/0.146/0.061, matching the observed range of raw on this dataset, whose 95th percentile is 0.382. Original labels are retained per row so the change is reversible. Removing the gate also makes the target a monotone function of a single scalar, which is easier for an ordinal model, so the gain over the uncalibrated run mixes class balance with target regularity and the two have not been separated. <sup>§</sup> GraphMind is complete on one backbone only. The Qwen3.5 row covers all 3,177 ideas; the deepseek-flash and gpt-5.6-luna rows were run on the graph ML slice alone, and on 724 and 714 of its 850 ideas, so their CV and time series cells are empty. Scoring Qwen3.5 on those same subsets moves it by at most 0.008 on any metric, so the Graph ML column can be read across the three backbones; the missing cells are not yet run rather than withheld.

**What the tables do and do not license.** Four observations survive. The backbone matters more than the prompt chain: in Table 2 the same five-prior single call moves from 0.342 to 0.459 BalAcc when Qwen3.5 is replaced by deepseek-flash, a larger move than any difference between the two prompt chains on a fixed backbone. GraphMind is the one place where that substitution does not help — on the graph ML slice it sits at 0.281, 0.289 and 0.351 BalAcc for Qwen3.5, deepseek-flash and gpt-5.6-luna, so the chain reorders the backbones rather than tracking them — and it is also the only prompt-based system in Table 2 that falls below a single direct call on its own backbone, 0.281 against 0.336 BalAcc on Qwen3.5 and 1.027 against 0.925 MAE. Its per-idea cost is the highest in the table at 476 seconds and 31,201 LLM calls for the 3,177 ideas. What the extra structure changes is which class the chain collapses onto: the same five retrieved priors are read as 4.33 supporting and 0.01 contrasting by Qwen3.5 and as 1.04 and 2.91 by deepseek-flash, and the resulting ratings collapse onto MODERATE in the first case and onto LOW/WEAK in the second.

The polarity stage, not the evidence, sets the level. The domain does not matter much: within any row of Table 2 the three domains differ by a few points, while the gap between the baselines and our model is above thirty. And on RINoBench the no-evidence condition, reported in the appendix, is numerically identical to the constant HIGH predictor on every metric, that is, without evidence the model emits a constant rather than a judgement. And more evidence is not uniformly better. DeepReviewer-7B, the one system in Table 3 fine-tuned on paper review rather than prompted, falls from 0.1425 to 0.1230 Macro-F1 when its evidence grows from five priors to the full RINoBench set, while Qwen3.5 rises from 0.2244 to 0.2477 over the same change; what the extra evidence buys is another 44 of the 277 ideas pushed into the WEAK class, from 200 to 244. At neither budget does it ever predict the top class, which holds 41.5% of the gold. The reading that fits both numbers is that a model trained to review whole papers treats a longer related-work section as evidence of less novelty, largely independently of what that work says — which is what scoring a target against a retrieved set read as a block, rather than against its content, permits. Three limits are equally visible. MAE is lowest for the systems that collapse toward the middle class, so it must be read with the prediction distribution rather than on its own. The high-novelty class remains the weakest of the four for our model in every domain, at 0.747–0.797 F1 against 0.881–0.905 for LOW. And the RINoBench transfer result depends on cuts fitted on the target train split; the same weights applied without them fall to Macro-F1 0.1243. One further caveat belongs with Table 3 specifically. Its accuracy column is the one metric on that benchmark that a constant predictor already leads, and our row reaches 0.4152 by moving the decision cuts, not by ordering the ideas better: Spearman correlation against the gold levels is 0.1896 either way. Accuracy on this benchmark is a statement about where the cuts sit, and we report it only so the row can be read beside the LLM rows, which are themselves mostly cut placement.

**Not yet in these tables.** The four ablation groups and the data-sensitivity sweep, covering EGRD training fraction at 10/25/50/75% and residual-type label noise at 10/20/30%, are implemented and packaged but have no retrieved results, so no ablation row appears. GraphMind now has a row in Table 2 on the Qwen3.5 backbone only; its deepseek-flash and gpt-5.6-luna runs cover the graph ML slice and leave six of the fifteen cells in that block empty. The DeepReviewer row of that table covers 2,975 of the 3,177 ideas: the remaining 202 produced no parseable rating even after a second turn, and since those are the generations that ran to the token cap rather than a random subset, the row is if anything optimistic. That row also takes 62% of its scores from that second turn, which departs from the single-round protocol of the original system. No table yet contains a domain-only row, that is, a model trained on one domain and evaluated on that domain’s slice of the same test split, which is the comparison that would show whether pooling the three domains helps at all. Every row carries a single seed; paired seeds 173/174/175 are required before two rows within a block are compared.

## 6 ABLATION AND PARAMETER SENSITIVITY

**Conditions under which these numbers were produced.** The sweeps share one configuration, varying a single parameter at a time: `deberta-v3-base` backbone, 8 epochs, batch size 2, learning rate  $1.5 \times 10^{-5}$  with an encoder ratio of 0.2, eight semantic units with slot competition enabled, and a fixed joint-prediction checkpoint supplying the encoder initialisation. The split is a date-ordered 17,804/3,815/3,816 partition of the 25,435 idea-level samples in the IdeaDelta; train/dev share the boundary month 202506 and dev/test share 202511, so the partition is not strictly month-disjoint and source-paper isolation has not been established. All sweep rows are single-seed (seed 173), so the overall shape of a curve is interpretable at the magnitudes reported, while a two- to three- $\sigma$  difference between adjacent points is not.

**How the per-domain figures are obtained.** Per-domain metrics are recomputed from the stored predictions rather than read from the trainer, and reconcile with the trainer’s own pooled figures to within  $5 \times 10^{-5}$  on all six metrics.

**Unfinished conditions.** Two points are missing:  $K=1$  and the 10% training fraction. The scheduler refused to dispatch them because the trainer source had changed after the twelve completed conditions were run, which would have made the results incomparable. Restoring the earlier trainer revision, or establishing that the change is an identity transformation under these defaults, is a pre-

Table 5: Component ablations, per domain. **Full** is the unablated EGRD checkpoint used in Table 2; High-F1 is also reported in Table 4. **Permuted prior evidence** permutes the retrieved prior texts; **permuted coverage** permutes only the joint coverage score while retaining the prior texts; **residuals zeroed** sets  $z_U$ ,  $z_I$  and  $z_C$  to zero simultaneously. Full and the ablated variants come from separate training batches. Acc is accuracy; BalAcc is balanced accuracy; Macro-F1 averages F1 over all four classes; QWK is quadratic weighted kappa; High-F1 is the F1 of the high-novelty class. Higher is better throughout. The permuted-prior-evidence row is recomputed from saved predictions on each identified domain; the permuted-coverage row is obtained by subtracting its recorded per-domain degradation from the Full row above; the residuals-zeroed row reports adjusted values.

| System                      | Acc           | BalAcc        | Macro-F1      | QWK           | High-F1      |
|-----------------------------|---------------|---------------|---------------|---------------|--------------|
| <i>Graph ML</i>             |               |               |               |               |              |
| Full                        | <b>0.7965</b> | <b>0.8024</b> | <b>0.7976</b> | <b>0.9076</b> | <b>0.797</b> |
| Permuted prior evidence     | 0.3471        | 0.3716        | 0.3532        | 0.3241        | 0.4420       |
| Permuted coverage           | 0.6649        | 0.6694        | 0.6605        | 0.5996        | 0.6630       |
| Residuals zeroed (adjusted) | 0.6371        | 0.6730        | 0.6554        | 0.7878        | 0.7479       |
| <i>Computer vision</i>      |               |               |               |               |              |
| Full                        | <b>0.7634</b> | <b>0.7804</b> | <b>0.7696</b> | <b>0.8823</b> | <b>0.748</b> |
| Permuted prior evidence     | 0.3694        | 0.3642        | 0.3586        | 0.3473        | 0.4078       |
| Permuted coverage           | 0.6368        | 0.6356        | 0.6192        | 0.5923        | 0.5880       |
| Residuals zeroed (adjusted) | 0.6332        | 0.5822        | 0.6414        | 0.7624        | 0.7397       |
| <i>Time series</i>          |               |               |               |               |              |
| Full                        | <b>0.7857</b> | <b>0.7831</b> | <b>0.7715</b> | <b>0.8963</b> | <b>0.757</b> |
| Permuted prior evidence     | 0.3697        | 0.3893        | 0.3714        | 0.3789        | 0.3952       |
| Permuted coverage           | 0.6252        | 0.6310        | 0.6164        | 0.5882        | 0.6196       |
| Residuals zeroed (adjusted) | 0.7196        | 0.7682        | 0.7332        | 0.8373        | 0.6748       |

Table 6: Per-domain seed variability for the five metrics defined in Table 5 and ordinal mean absolute error (MAE). Entries are sample standard deviations across seeds 173, 174 and 175 with identical data, code and configuration. These estimates provide context for Tables 7–9; three runs are insufficient for precise significance claims.

| Domain          | Acc    | BalAcc | Macro-F1 | QWK    | MAE    | High-F1 |
|-----------------|--------|--------|----------|--------|--------|---------|
| Graph ML        | 0.0066 | 0.0078 | 0.0050   | 0.0048 | 0.0067 | 0.0022  |
| Computer vision | 0.0040 | 0.0048 | 0.0021   | 0.0032 | 0.0027 | 0.0037  |
| Time series     | 0.0050 | 0.0043 | 0.0045   | 0.0031 | 0.0051 | 0.0069  |

requisite for adding them. Neither point affects the conclusions drawn from the completed conditions, but both would extend the left end of their curves.

## 7 WHAT THE REPRESENTATION ORGANISES

The grade our model predicts and the representation it predicts it from are not the same object, and the second is the more informative one to inspect. This section reads six stage representations out of a single forward pass of one frozen checkpoint over the 3,816 test ideas: the target encoding, the covered pathway  $z_C$ , the two residual pathways  $z_U$  and  $z_I$ , the global feature context, and the fused vector the prediction heads receive.

**Choosing a target the input scalars cannot reach.** Two quantities enter the model as features and between them carry most of the four-way novelty grade: the joint coverage of the retrieved priors and the number of admissible candidates. A multinomial probe on those two scalars and their low-order interactions reaches 0.681 accuracy on the grade, against a 0.329 majority baseline and the model’s own 0.778. Any structure we find by probing for the grade would therefore be reporting, in large part, on those two numbers.

Table 7: Sensitivity to the evidence budget  $K$ , per domain.  $K$  is a hard padding dimension: every sample is padded to exactly  $K$  slots before encoding, so nominal  $K$  and the evidence actually consumed diverge once  $K$  exceeds the number of candidates an idea carries. Averaged over the evaluation set the consumed counts are 2.00, 2.80, 3.41, 3.87, 4.19 and 4.54 priors for  $K = 2, 3, 4, 5, 6, 8$ , and the curve should be read against those rather than against  $K$ . Only  $K$  is varied; upstream pair and joint products, the encoder initialisation and the evaluation files are shared across all rows, and  $K=5$  is the default. The shape reproduces independently in all three domains: a steep rise from  $K=2$  to  $K=4$ , then a plateau, with the knee at roughly 3.4 consumed priors. The plateau is a property of the data rather than the model, since only 32% of ideas carry six or more candidates and 13.5% carry eight or more, so raising  $K$  from 5 to 8 adds 0.67 priors on average. MAE improves monotonically over the rising part of the curve in every domain.  $K=1$  was not run.

| $K$                    | Acc    | BalAcc | Macro-F1 | QWK    | MAE ↓  | High-F1 |
|------------------------|--------|--------|----------|--------|--------|---------|
| <i>Graph ML</i>        |        |        |          |        |        |         |
| 2                      | 0.7306 | 0.7466 | 0.7383   | 0.8779 | 0.2706 | 0.8103  |
| 3                      | 0.7753 | 0.7906 | 0.7786   | 0.9009 | 0.2247 | 0.8127  |
| 4                      | 0.8059 | 0.8106 | 0.8077   | 0.9099 | 0.1941 | 0.8235  |
| 5                      | 0.7988 | 0.8084 | 0.8010   | 0.9094 | 0.2012 | 0.8155  |
| 6                      | 0.7824 | 0.7862 | 0.7820   | 0.9007 | 0.2176 | 0.7867  |
| 8                      | 0.7929 | 0.7963 | 0.7950   | 0.9037 | 0.2071 | 0.8000  |
| <i>Computer vision</i> |        |        |          |        |        |         |
| 2                      | 0.7159 | 0.7272 | 0.7188   | 0.8561 | 0.2885 | 0.7743  |
| 3                      | 0.7476 | 0.7721 | 0.7529   | 0.8750 | 0.2577 | 0.7422  |
| 4                      | 0.7722 | 0.7743 | 0.7761   | 0.8818 | 0.2304 | 0.7653  |
| 5                      | 0.7608 | 0.7790 | 0.7669   | 0.8818 | 0.2419 | 0.7448  |
| 6                      | 0.7529 | 0.7650 | 0.7583   | 0.8756 | 0.2498 | 0.7500  |
| 8                      | 0.7722 | 0.7828 | 0.7774   | 0.8836 | 0.2313 | 0.7540  |
| <i>Time series</i>     |        |        |          |        |        |         |
| 2                      | 0.7261 | 0.7407 | 0.7249   | 0.8679 | 0.2782 | 0.7697  |
| 3                      | 0.7765 | 0.7844 | 0.7637   | 0.8917 | 0.2294 | 0.7622  |
| 4                      | 0.7891 | 0.7905 | 0.7805   | 0.8960 | 0.2134 | 0.7823  |
| 5                      | 0.7891 | 0.7906 | 0.7760   | 0.8975 | 0.2151 | 0.7697  |
| 6                      | 0.7899 | 0.7857 | 0.7761   | 0.8954 | 0.2143 | 0.7604  |
| 8                      | 0.7824 | 0.7805 | 0.7701   | 0.8931 | 0.2210 | 0.7516  |

The residual type does not have this problem. It is an eleven-way label naming what kind of contribution remains once the priors are accounted for — a new mechanism, a non-trivial combination, a minor variant, a new evaluation protocol, and so on — and it is not a function of how much coverage there is. The same two-scalar probe reaches only 0.368 on it against a 0.275 majority, while the model’s own head reaches 0.651. The gap of twenty-eight points is the part that had to be read out of the text, and it is what the rest of this section examines. Figure 2 collects the result.

**The types are laid out, and the layout predicts the model’s own errors.** Projecting the eleven centroids to two dimensions recovers 61.9% of their spread, so the arrangement is close to planar. Its first axis correlates at  $-0.905$  with the mean novelty grade of each type across the eleven classes, which places the conceptual and task-defining contributions at one end and minor variants and recombinations at the other.

The arrangement is not decoration: it predicts where the model goes wrong. Over the 55 type pairs, the cosine between two centroids correlates at  $r = 0.469$  with how often the model swaps those two types. The model never mistakes a new data distribution for a new theoretical concept, the most distant pair in the map, and its single most frequent confusion, minor variant against non-trivial combination, is one of its closest.

That correlation could be a restatement of the novelty axis, since types of similar novelty would be confusable for that reason alone. It is not. Partialling out the difference in mean novelty between the two types raises the correlation to 0.600 rather than removing it. The clearest case is NEW\_MECHANISM against NEW\_TRAINING\_SIGNAL, whose mean grades differ by 0.05 and which

Table 8: Sensitivity to the amount of EGRD training data, per domain. Training ideas are sub-sampled stratified by domain and novelty label, and the four fractions are nested prefixes of one deterministic random order ( $25\% \subset 50\% \subset 75\% \subset 100\%$ ), so consecutive rows add samples without reordering what came before; the absolute counts are 4,451, 8,902, 13,353 and 17,804 training ideas. Development and test sets are fixed. Epochs are held at 8, so a row with less data also takes fewer optimisation steps, and this is a learning curve under a fixed epoch count rather than under a fixed compute budget; separating the two requires an equal-update-budget control that was not run. Upstream pair and joint products are held fixed and were fitted on the full training data, so the 25% row does not license a claim about the annotation efficiency of the whole pipeline. Macro-F1 increases monotonically with more data in Graph ML and time series. Computer vision reaches its highest Macro-F1 at 50% (0.7788), compared with 0.7669 at 100%. The 10% condition was not run.

| Fraction               | Acc    | BalAcc | Macro-F1 | QWK    | MAE ↓  | High-F1 |
|------------------------|--------|--------|----------|--------|--------|---------|
| <i>Graph ML</i>        |        |        |          |        |        |         |
| 25%                    | 0.7600 | 0.7664 | 0.7627   | 0.8865 | 0.2424 | 0.7829  |
| 50%                    | 0.7871 | 0.7893 | 0.7871   | 0.9021 | 0.2129 | 0.7838  |
| 75%                    | 0.7894 | 0.7927 | 0.7897   | 0.9032 | 0.2106 | 0.7907  |
| 100%                   | 0.7988 | 0.8084 | 0.8010   | 0.9094 | 0.2012 | 0.8155  |
| <i>Computer vision</i> |        |        |          |        |        |         |
| 25%                    | 0.7634 | 0.7710 | 0.7687   | 0.8756 | 0.2410 | 0.7494  |
| 50%                    | 0.7740 | 0.7825 | 0.7788   | 0.8839 | 0.2287 | 0.7537  |
| 75%                    | 0.7661 | 0.7769 | 0.7727   | 0.8799 | 0.2375 | 0.7581  |
| 100%                   | 0.7608 | 0.7790 | 0.7669   | 0.8818 | 0.2419 | 0.7448  |
| <i>Time series</i>     |        |        |          |        |        |         |
| 25%                    | 0.7731 | 0.7680 | 0.7573   | 0.8862 | 0.2319 | 0.7188  |
| 50%                    | 0.7790 | 0.7773 | 0.7662   | 0.8911 | 0.2252 | 0.7492  |
| 75%                    | 0.7849 | 0.7817 | 0.7714   | 0.8917 | 0.2202 | 0.7452  |
| 100%                   | 0.7891 | 0.7906 | 0.7760   | 0.8975 | 0.2151 | 0.7697  |

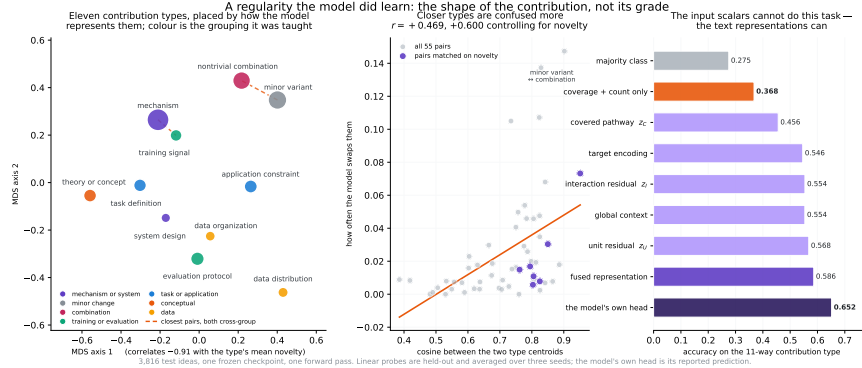


Figure 2: Left: the eleven contribution types, placed by classical multidimensional scaling on the cosine distances between their centroids in the fused representation; two dimensions carry 61.9% of the centroid spread, and point area is class frequency. Colour is the seven-way coarse grouping the model is supervised with, so same-coloured points sitting apart means the layout is not a replay of that supervision; the dashed segments mark the two closest pairs in the whole map, both of which cross groups. Centre: each of the 55 type pairs, its centroid cosine against how often the model swaps the two. Right: held-out accuracy of a linear probe for the eleven-way type, averaged over three seeds.

novelty therefore cannot separate at all: they are the closest pair in the entire map at cosine 0.951, and among the novelty-matched pairs they are also the most often confused.

Table 9: Sensitivity to encoder input length on the three domain test splits. All entries are absolute values of the five metrics defined in Table 5 and ordinal MAE, recomputed from the saved test predictions. The default is 384; all settings use seed 173 and the same test ideas and labels. Graph ML has its highest Macro-F1 at 384, whereas computer vision and time series have their highest Macro-F1 at 128. No domain exhibits a monotonic Macro-F1 improvement with length. These single-seed results do not establish an optimal length; runtime and token-length distributions were not measured.

| max_length             | Acc    | BalAcc | Macro-F1 | QWK    | MAE ↓  | High-F1 |
|------------------------|--------|--------|----------|--------|--------|---------|
| <i>Graph ML</i>        |        |        |          |        |        |         |
| 128                    | 0.7953 | 0.8028 | 0.7974   | 0.9018 | 0.2082 | 0.8065  |
| 192                    | 0.7918 | 0.7966 | 0.7917   | 0.9056 | 0.2082 | 0.7960  |
| 256                    | 0.7953 | 0.8084 | 0.7978   | 0.9090 | 0.2047 | 0.8228  |
| <b>384</b>             | 0.7988 | 0.8084 | 0.8010   | 0.9094 | 0.2012 | 0.8155  |
| <i>Computer vision</i> |        |        |          |        |        |         |
| 128                    | 0.7766 | 0.7881 | 0.7829   | 0.8841 | 0.2278 | 0.7816  |
| 192                    | 0.7634 | 0.7820 | 0.7687   | 0.8832 | 0.2392 | 0.7653  |
| 256                    | 0.7599 | 0.7828 | 0.7662   | 0.8831 | 0.2427 | 0.7613  |
| <b>384</b>             | 0.7608 | 0.7790 | 0.7669   | 0.8818 | 0.2419 | 0.7448  |
| <i>Time series</i>     |        |        |          |        |        |         |
| 128                    | 0.7941 | 0.7966 | 0.7840   | 0.9003 | 0.2084 | 0.7799  |
| 192                    | 0.7840 | 0.7783 | 0.7696   | 0.8948 | 0.2193 | 0.7582  |
| 256                    | 0.7941 | 0.7998 | 0.7813   | 0.9010 | 0.2101 | 0.7768  |
| <b>384</b>             | 0.7891 | 0.7906 | 0.7760   | 0.8975 | 0.2151 | 0.7697  |

**The layout disagrees with the grouping the model was taught.** Training supervises a seven-way coarse grouping alongside the fine label, so the layout could be that grouping made visible. It is not. Pairs inside a coarse group sit at mean cosine 0.783 and pairs across groups at 0.709, a separation far smaller than the spread within either set, and the four closest pairs in the map all cross groups. The model places MINOR\_VARIANT beside NONTRIVIAL\_COMBINATION at 0.902 although the supervision assigns them to different coarse types, and it pulls NEW\_MECHANISM away from NEW\_SYSTEM\_DESIGN to 0.798 although the supervision groups them together. Both departures read as the more natural organisation: a minor variant and a recombination are two degrees of reusing what exists, whereas a new mechanism and a new system design are different kinds of contribution. We report this as an observation about the representation, not as evidence that the taxonomy should be redrawn; with only four same-group pairs available the comparison of the two means is descriptive.

**Where the knowledge sits.** Probing each stage separately for the eleven-way type puts the covered pathway lowest at 0.456, the target encoding at 0.546, the interaction residual at 0.554, the global context at 0.554, the unit residual highest among the single pathways at 0.568, and the fused vector at 0.586. Two readings follow. Every text-derived representation is well clear of the 0.368 the input scalars reach, so the contribution type is genuinely carried by the encoded text rather than by the retrieval statistics. And the target encoding alone already reaches 0.546, which says that much of the type is legible from how the idea is phrased — a target that says it combines two things announces its own category — so the evidence pathways add less here than the gap to the model’s 0.651 head might suggest.

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## A APPENDIX

### A.1 DATASET CONSTRUCTION

**Sources and idea extraction.** Papers are drawn from arXiv in three areas—graph machine learning, computer vision, and time series. Selection prefers work that was subsequently published and, among that, papers at major venues. From each paper’s title, abstract and introduction an LLM extracts innovation claims and assigns each an innovation aspect; a second LLM pass discards claims that are not innovation claims at all—open-source and reproducibility statements, bare state-of-the-art assertions, and engineering deliverables—so that later stages judge substantive claims rather than boilerplate.

**Candidate priors.** Idea texts are embedded and indexed for top- $K$  nearest-neighbour recall, and candidate relations are formed under a hybrid rule: high similarity is accepted directly, low similarity is rejected directly, and the uncertain band is passed to an LLM. The base build pairs ideas of the same type only, which would record an idea whose closest relative is of another type—a theoretical contribution nearest to a method—as isolated; a cross-type pass recovers these. At most  $K = 5$  ranked priors are retained per target. Pairing stays within a source domain, and no cross-domain pair is labelled, so the three single-domain subsets and their union address the same ideas under identifiers namespaced by domain. The assembled dataset passes its own validation checks with no duplicate pair keys, no duplicate novelty targets, no invalid pair labels, no target left without a pair, and no temporal violation.

**Annotation.** All annotation stages use the same judge, `deepseek-v4-pro`. It labels pair coverage over the four classes, with the ambiguous band—related but not covering, partial cover, large cover, and high-similarity pairs the graph left unconnected—re-judged rather than accepted from the similarity rule. It then estimates, for each target, the substantive residual that survives the retained evidence, the sufficiency of that evidence, and the extent to which the claimed difference is surface-level. Idea-level grades follow by applying Eq. (1) to those quantities. Labels are model-generated; each stage was reviewed by sampled manual inspection.

**Splits and balancing.** Splits are date-ordered rather than random, so that no training target is dated after an evaluation target. The cuts fall at 2025-02 and 2025-11, giving 59,269/16,436/17,171 pairs and 13,815/3,944/3,352 ideas. The cut months are shared at the boundary, so the partition is ordered but not strictly month-disjoint, and isolation at the level of source papers has not been established. A balanced training variant resamples the pair training set to 1,000 examples per coverage class, at most five copies of any example and marked as duplicates; dev and test keep their natural distribution and are never resampled. Pair coverage labels are skewed in the natural distribution: of 92,876 pairs, 41,765 are not covering, 38,775 related but not covering, 9,582 partial cover, and 2,754 large cover.

## A.2 VALIDATION OF RESEARCH NOVELTY LABELS

Our novelty labels assess the substantive contribution of a target idea beyond its retrieved prior work. We validate this interpretation through complementary experiments. First, we examine how well metadata and retrieved prior content predict the existing labels. We then modify the evidence and repeat annotation to test whether novelty decreases when the target contribution is already covered and remains stable when its substantive contribution is unchanged.

**Prediction from metadata.** We first test whether simple metadata can account for the novelty labels without describing the technical content of prior work. We consider seven variables: character length, word count, publication year, three research-area attributes, and the number of retrieved priors. Over all 21,111 ideas, we measure their individual associations with the novelty grade using Spearman  $\rho$  for continuous variables and  $\varepsilon^2$  for categorical variables. We set the thresholds in advance at  $|\rho| \geq 0.20$  and  $\varepsilon^2 \geq 0.06$ .

Text length, word count, and publication year show weak associations ( $\rho = +0.049, +0.058$ , and  $-0.130$ , respectively), while the three research-area attributes have  $\varepsilon^2 \leq 0.033$ . All remain below the corresponding thresholds. The number of priors and their mean coverage are negatively correlated with novelty ( $\rho = -0.286$  and  $-0.471$ , respectively). The negative association with coverage is consistent with the intended meaning of the labels: contributions already covered by prior work receive lower novelty grades.

Individual associations may overlook information available from several variables together. We therefore train a classifier on all seven metadata variables to predict the four novelty grades. We use the date-ordered split in Section 3.3, with 13,815 training ideas and 3,352 test ideas, and evaluate quadratic weighted kappa (QWK), which accounts for the ordering of the grades. For comparison, we train the same classifier using only mean prior coverage. QWK increases from 0.227 with metadata to 0.539 with coverage alone. Coverage thus predicts the labels more accurately than the metadata considered here, supporting the relevance of the relationship between the target and prior work.

**Prediction with retrieved and random priors.** We next examine whether the specific retrieved content matters for label prediction. In the TRUE condition, each target is paired with its five retrieved priors. In the RANDOM condition, these are replaced by five ideas sampled uniformly from those published before the target. The two conditions use the same number of priors and total evidence text volume, and all priors precede the target. We hold the model, hyperparameters, date-ordered split, and random seeds fixed. Models are trained separately from scratch and evaluated on the same held-out targets against their original labels.

Across the three seeds, QWK decreases from 0.196 with retrieved priors to  $0.005 \pm 0.021$  with random priors. The gap of 0.192 is about an order of magnitude larger than the between-seed standard deviation of 0.015. Replacing retrieved priors with random earlier work therefore reduces

prediction to near-chance agreement. This comparison supports the importance of the retrieved content, beyond the amount of evidence supplied.

**Response to evidence covering the target contribution.** The prediction experiments keep the labels fixed. We now test whether the annotation procedure responds appropriately when the evidence changes. If an earlier study already contains the target contribution, the assigned novelty should decrease. We sample 30 targets across the four grades and construct one synthetic prior for each target. This prior describes the same technical contribution and is dated before the target. After adding it to the evidence set, we repeat the complete annotation procedure: pair coverage labeling, residual contribution estimation, and grading according to Eq. (1). Synthetic priors have dedicated identifiers and are marked `synthetic_control`. They do not refer to real papers, authors, or DOIs and are excluded from the released dataset.

Each modified target is compared with five independent annotations of its original evidence set to account for variation across repeated annotations. We evaluate recognition of the added prior as large coverage, changes in joint coverage and residual contribution, and decreases in the novelty score and grade. Ties are excluded from the exact sign test.

The added prior is labeled large coverage in all 29 comparable cases. Joint coverage increases, while residual contribution and novelty score decrease. The novelty grade decreases for 82.8% of targets (sign test  $p = 2.2 \times 10^{-7}$ ). These results show that the annotation procedure lowers novelty when the evidence covers the target contribution.

**Stability under changes that preserve the contribution.** A novelty judgment should also remain stable when the evidence changes without altering the substantive contribution of the target. For the same 30 targets, we reorder the priors, duplicate existing priors, or add unrelated papers. The novelty grades before and after these changes show high agreement:  $\kappa = 0.970, 0.902$ , and  $1.000$ , respectively. These results complement the coverage experiment: the judgments respond to evidence that reduces the remaining contribution and remain comparatively stable under changes that preserve it.

**Evaluation scope.** The metadata analysis covers 21,111 ideas, and the prediction experiments use 13,815/3,944/3,352 training, validation, and test examples ordered by date. The evidence modification experiments cover 30 targets, with 150 baseline repetitions, 30 coverage comparisons, and 160 stability comparisons. Sampling uses a fixed seed and groups ideas by source paper. Each modified example is compared with its own original version, and confidence intervals for component-level changes use 2,000 bootstrap samples clustered by target.

## B METHOD DETAILS

### B.1 HISTORICAL COVERAGE MODELS

**Pair coverage and evidence selection.** For each target–prior pair  $(x, p_k)$ , the pair coverage model estimates how much of the target the prior accounts for. It outputs a score  $s_k \in [0, 1]$  and a distribution over four classes: unrelated, related but uncovered, partially covered, and largely covered. A cross-encoder predicts relatedness  $a_k$ , coverage conditional on relatedness  $b_k$ , and large coverage conditional on coverage  $d_k$ . These probabilities define the class distribution

$$\pi_k^{\text{pair}} = [1 - a_k, a_k(1 - b_k), a_k b_k(1 - d_k), a_k b_k d_k]. \quad (8)$$

With classes indexed from 0 to 3, the coverage score is  $s_k = \frac{1}{2}\pi_{k,2}^{\text{pair}} + \pi_{k,3}^{\text{pair}}$ . The reference pipeline ranks temporally admissible candidates by  $s_k + 0.02\ell_k/3$ , where  $\ell_k$  is the predicted ordinal coverage label, and retains at most  $K$  priors. The reported IdeaDelta checkpoint instead uses the ordering and pair annotations supplied in the dataset ranking records, as detailed under the evidence-source protocol below. In both cases, the retained priors and their pair coverage features form the evidence inputs to the subsequent stages.

**Joint coverage.** Different priors may cover different parts of the target. We therefore estimate their joint coverage with a multiple-instance model, treating each pair  $(x, p_k)$  as an instance and the

retained evidence set as a bag (Ilse et al., 2018). Let  $n = |\mathcal{E}_x|$  denote the number of retained priors. A shared cross-encoder maps each pair to a representation  $\mathbf{h}_k^J$ , from which the model computes attention weights  $\beta_k$  and internal evidence scores  $e_k$ :

$$\beta_k = \text{softmax}_k(\mathbf{w}_a^\top \mathbf{h}_k^J + b_a), \quad e_k = \sigma(\mathbf{w}_e^\top \mathbf{h}_k^J + b_e). \quad (9)$$

For a nonempty evidence set, noisy-OR pools the evidence scores as

$$e_U = 1 - \prod_{k=1}^n (1 - e_k). \quad (10)$$

The model summarizes these scores by their noisy-OR aggregate, maximum, mean, and the fraction of the  $K$  evidence slots that are filled. It concatenates these statistics with attention-pooled and elementwise maximum-pooled pair representations:

$$\mathbf{v}_E = \left[ e_U, \max_k e_k, \frac{1}{n} \sum_{k=1}^n e_k, \frac{n}{K} \right], \quad (11)$$

$$\mathbf{z}_J = f_J \left( \left[ \sum_{k=1}^n \beta_k \mathbf{h}_k^J; \max_k \mathbf{h}_k^J; \mathbf{v}_E \right] \right). \quad (12)$$

Here  $f_J$  is an MLP. Attention pooling aggregates evidence across priors, while maximum pooling retains the strongest feature values. A hierarchical head maps  $\mathbf{z}_J$  to three probabilities: any coverage, substantial coverage conditional on any coverage, and mostly covered conditional on substantial coverage. Using the factorization in Eq. (8), these probabilities yield  $\pi^J$  over four joint-coverage classes: *not*, *weakly*, *partially*, and *mostly* covered. The scalar joint-coverage estimate is  $C_J = \sum_{\ell=0}^3 \mu_\ell \pi_\ell^J$ , where  $\mu_\ell$  is the mean annotated coverage score for class  $\ell$  on the training split. Both  $C_J$  and  $\pi^J$  are supplied as fixed inputs to the residual and novelty modules. The noisy-OR aggregate  $e_U$  contributes one of the four statistics in  $\mathbf{v}_E$ ; the joint-coverage prediction is learned from the full bag representation.

## B.2 RESIDUAL REPRESENTATIONS AND POOLING

**Pair encoding.** For each target–prior pair  $(x, p_k)$ , the residual cross-encoder produces token states  $\mathbf{H}_k \in \mathbb{R}^{L \times d}$ , where  $L$  is the input sequence length. We apply masked mean pooling to obtain the pair representation and augment it with projected pair coverage features:

$$\mathbf{v}_k = \text{MeanPool}(\mathbf{H}_k), \quad \tilde{\mathbf{v}}_k = \mathbf{v}_k + f_P(\mathbf{f}_k), \quad (13)$$

where  $f_P$  consists of a linear map, layer normalization, GELU, and dropout. The residual encoder shares parameters across all retained priors. To construct the target token bank, we average target-token states at matching positions across valid pairs. Masked mean pooling over this bank yields the target representation  $\bar{\mathbf{h}}_x$  used in residual fusion.

**Coverage reconstruction.** To supervise the unit coverage estimates, we aggregate them into a scalar joint-coverage prediction. The pooling weights depend on both the semantic representation and the activity of each unit:

$$\hat{C}_U = \sum_{m=1}^M \omega_m c_m, \quad \omega_m = \frac{v_m \exp(\mathbf{w}_c^\top \mathbf{u}_m + b_c)}{\sum_{r=1}^M v_r \exp(\mathbf{w}_c^\top \mathbf{u}_r + b_c)}. \quad (14)$$

The resulting estimate  $\hat{C}_U$  is supervised by annotated coverage and the fixed Stage A prediction. This aggregate supervision constrains unit coverage but does not uniquely determine the latent support matrix. We use noisy-OR as a differentiable rule for aggregating support; its use does not establish that support from different priors is statistically independent.

**Interaction encoding and attention.** For each unit pair  $i < j$ , an MLP  $f_I$  encodes the two units and their elementwise comparisons into an interaction representation. A sigmoid head then predicts the interaction strength:

$$t_{ij} = f_I([\mathbf{u}_i; \mathbf{u}_j; \mathbf{u}_i \odot \mathbf{u}_j; |\mathbf{u}_i - \mathbf{u}_j|]), \quad g_{ij} = \sigma(\mathbf{w}_g^\top t_{ij} + b_g). \quad (15)$$

Using the interaction residual weights  $r_{ij}$  defined in Stage B, we compute attention over all  $\binom{M}{2}$  unit pairs:

$$\eta_{ij} = \text{softmax}_{i < j} (\mathbf{w}_I^\top \mathbf{t}_{ij} + b_I + \log(r_{ij} + \varepsilon)), \quad \mathbf{z}_I = \sum_{i < j} \eta_{ij} r_{ij} \mathbf{t}_{ij}. \quad (16)$$

The logarithmic bias directs attention toward pairs with larger residual weights. Multiplication by  $r_{ij}$  during pooling also preserves the scale of the residual weights, allowing weak residuals to contribute less even after attention normalization.

### B.3 RESIDUAL FUSION, PREDICTION HEADS, AND DECISION RULES

**Context gate.** The fusion gate conditions the contribution of global context on the two residual representations and the uncertainty of the joint-coverage prediction. Let  $H_J$  denote the normalized entropy of  $\pi^J$ . The gate is

$$\gamma = \sigma(f_\gamma([\mathbf{z}_U; \mathbf{z}_I; \mathbf{z}_G; H_J])). \quad (17)$$

The gated context  $\gamma \mathbf{z}_G$  enters residual fusion alongside  $\mathbf{z}_U$  and  $\mathbf{z}_I$ . Global context  $\mathbf{z}_G$  also enters the final shared representation directly. Thus,  $\gamma$  controls the context contribution within the residual branch, while the direct context path remains available to the prediction heads.

**Auxiliary predictions.** The auxiliary heads use the shared representation  $\mathbf{h}$  to predict substantive difference  $\hat{d}$ , surface change  $\hat{s}$ , evidence sufficiency  $\hat{e}$ , and residual-type probabilities  $\pi^T$ . The first three outputs are sigmoid scores; the type prediction uses the hierarchy described below.

**Residual types.** The type hierarchy distinguishes a minor variant from six substantive families: conceptual innovation, mechanism or system, task or application, training or evaluation, data-related innovation, and combination. Conditional fine heads yield eleven types in total. The minor/substantive gate additionally reads  $\hat{d}$ ,  $\hat{s}$ , and  $\max_{i < j} r_{ij}$ . The probability assigned to the combination type is multiplied by  $0.70 + 0.30\sigma(f_{\text{comb}}(\mathbf{z}_I))$  before the hierarchical distribution is renormalized. The final type distribution is a convex mixture of the hierarchical and flat predictions, with flat weight 0.15.

**Novelty branches.** All novelty branches receive the prediction features

$$\xi = [\mathbf{h}; \hat{d}; \hat{s}; \hat{e}; \pi^T; C_J; \pi^J]. \quad (18)$$

We encode LOW, WEAK, MODERATE, and HIGH as 0, 1, 2, and 3, respectively. A direct linear classifier and a two-layer MLP classifier map  $\xi$  to class distributions  $\mathbf{p}^D$  and  $\mathbf{p}^R$ .

The ordinal branch models cumulative probabilities of reaching each grade threshold. It produces three sigmoid gates  $g_1, g_2, g_3$  from  $\xi$  and defines

$$q_1 = g_1, \quad q_2 = g_1 g_2, \quad q_3 = g_1 g_2 g_3, \quad \mathbf{p}^O = [1 - q_1, q_1 - q_2, q_2 - q_3, q_3]. \quad (19)$$

Here  $q_\ell$  estimates the probability that  $y \geq \ell$ , with  $q_1 \geq q_2 \geq q_3$  by construction.

**Probability fusion.** We aggregate the three branch distributions through a convex mixture:

$$\mathbf{p} = \lambda_D \mathbf{p}^D + \lambda_O \mathbf{p}^O + \lambda_R \mathbf{p}^R, \quad \lambda_D + \lambda_O + \lambda_R = 1, \quad \lambda_D, \lambda_O, \lambda_R \geq 0. \quad (20)$$

The fused distribution provides the initial grade for the refinement below.

**Moderate-to-high refinement.** A separate binary head  $h_{MH} = \sigma(f_{MH}(\xi))$  scores the distinction between moderate and high novelty. At inference, let  $y_0 = \arg \max_\ell p_\ell$  for the mixture in Eq. (20), and define

$$h_{\text{high}} = \alpha_D p_3^D + \alpha_O p_3^O + \alpha_{MH} h_{MH}, \quad \hat{y} = \begin{cases} 3, & y_0 = 2 \text{ and } h_{\text{high}} \geq \theta_{MH}, \\ y_0, & \text{otherwise.} \end{cases} \quad (21)$$

where  $\alpha_D, \alpha_O, \alpha_{MH} \geq 0$  and  $\alpha_D + \alpha_O + \alpha_{MH} = 1$ . The refinement applies a threshold to this weighted score and does not require separate agreement from all three heads. For example, an initial MODERATE prediction is promoted to HIGH if the combined score reaches  $\theta_{MH}$ , even if the individual heads favor different grades. The distribution-mixture weights and refinement threshold are selected on development data and fixed for testing. Appendix B.4 reports the implementation values.

**Continuous outputs.** A separate sigmoid head predicts a continuous novelty score. At inference, novelty grades are determined by probability fusion and the moderate-to-high refinement; the continuous score output does not enter that decision rule. The continuous score head and the substantive-difference head are separate readouts; the latter is used in the blockwise RINoBench evaluation.

#### B.4 TRAINING AND IMPLEMENTATION DETAILS

This section supplements the training objective in Eq. (7) with the loss definitions and implementation settings for the prediction modules of Appendix B.3 and the residual decomposition of Appendix B.2.

**Supervision.** The classification group  $\mathcal{L}_{\text{nov}}$  trains the direct and MLP classifiers with class-weighted cross-entropy, the ordinal outputs against  $\mathbf{1}[y \geq \ell]$ ,  $\ell = 1, 2, 3$ , with weighted binary cross-entropy, and the fused distribution with a negative log-likelihood. The moderate-to-high head sees only moderate and high examples and adds a positive-negative margin to its binary cross-entropy, weighting moderate examples scored as high more heavily. The type group  $\mathcal{L}_{\text{type}}$  combines the hierarchy’s conditional losses, a flat classification loss, binary supervision for nontrivial combinations, and consistency penalties tying the combination branch to the type predictions. The separate novelty score regresses  $\nu^*$ , the continuous annotation target defined by  $\text{raw}(x)$  in Eq. (1) and formed from the annotated coverage, substantive difference, evidence sufficiency, and surface change of the target. It is the quantity the dataset thresholds into the four grades.

**Scalar heads: level and ranking terms.** Let  $\ell_\delta$  denote smooth- $L_1$  loss and  $w_x$  the example weight. For  $a \in \mathcal{A}$ , the level loss is  $\mathcal{L}_a = \sum_x w_x \ell_\delta(\hat{a}_x - a_x^*) / \sum_x w_x$ , where  $a_x^*$  is the annotation and  $\hat{a}_x$  the prediction. The corresponding ranking term takes within-minibatch pairs separated by  $|a_x^* - a_{x'}^*| \geq 0.05$  and contributes  $[0.02 - \text{sgn}(a_x^* - a_{x'}^*)(\hat{a}_x - \hat{a}_{x'})]_+$ , weighted by  $\sqrt{w_x w_{x'}}$ ; it is dropped when a minibatch holds no eligible pair. The ranking terms of  $\Delta$ ,  $S$ ,  $E$ , and  $C$  are summed into  $\mathcal{L}_{\text{rank}}$  with the internal weights of Table 10; the novelty score’s ranking term is carried inside  $\mathcal{L}_\nu$  instead, where it takes most of the weight because the grade is read off the score by thresholding and therefore depends only on order.

The ranking terms are not redundant with the level losses. At the transition  $\delta$  and target spread used here the level losses operate in their  $L_1$  regime, whose optimum is the median of the target, and a head supervised by the level loss alone drifts towards emitting a near-constant value; the evidence head is the clearest case. Ranking is scale-free and counteracts this.

**Coverage term.** Beyond its level loss against the annotated joint value  $C^*$ ,  $\mathcal{L}_C$  adds a consistency loss against the upstream score  $C_J$  at relative weight 0.2, reweighted by  $1 - H_J$  so that examples with uncertain upstream coverage count for less. No observed unit-level support matrix is supplied as a target.

**Decomposition regularization and variance floor.**  $\mathcal{L}_{\text{dec}}$  combines three terms. Attention regularization penalizes the squared entries of  $\tilde{\alpha}\tilde{\alpha}^\top - I$  for the row-normalized attention matrix  $\tilde{\alpha}$ , plus half the mean off-diagonal cosine similarity. Query diversity penalizes the mean squared cosine similarity between distinct queries. A usage penalty  $D_{\text{KL}}(\rho||\mathbf{u})$ , with  $\rho_m = \sum_j B_{mj} / \sum_{r,j} B_{rj}$  and  $\mathbf{u}$  uniform over the  $M$  units, keeps units from going unused. Their relative weights are given in Table 10.

The variance floor is a separate term rather than a component of  $\mathcal{L}_{\text{dec}}$ , and enters Eq. (7) under its own coefficient:

$$\mathcal{L}_{\text{var}} = \mathbb{E}_x [[\nu - \text{Var}_m(c_{x,m})]_+ / \nu]. \quad (22)$$

It acts on the unit coverages rather than on the token assignments, and keeps the  $c_m$  within an example from coinciding. The reported configuration uses softmax competition with dot-product scores.

**Curriculum and feature perturbation.** For a randomly sampled fraction of training examples the joint-coverage features, together with the associated probability and entropy features, are replaced by values derived from annotated joint coverage, with probability 0.50 in epoch 1, 0.20 in epoch

Table 10: Residual-stage settings for the reported IdeaDelta checkpoint. The checkpoint was selected on development data at epoch 5 of 8.

| Setting                                                       | Value                                   |
|---------------------------------------------------------------|-----------------------------------------|
| Training batch size / evaluation batch size                   | 2 / 8                                   |
| Head learning rate / encoder learning rate                    | $1.5 \times 10^{-5} / 3 \times 10^{-6}$ |
| Weight decay / dropout                                        | 0.01 / 0.15                             |
| Smooth- $L_1$ transition $\delta$                             | 0.25                                    |
| $\lambda_\Delta / \lambda_S / \lambda_E$                      | 1.0 / 0.6 / 0.8                         |
| $\lambda_C / \lambda_\nu$                                     | 0.3 / 0.6                               |
| $\lambda_{\text{nov}} / \lambda_{\text{type}}$                | 0.8 / 1.0                               |
| $\lambda_{\text{rank}}$                                       | 0.5                                     |
| $\lambda_{\text{dec}} / \lambda_{\text{var}}$                 | 0.05 / 0.02                             |
| Within $\mathcal{L}_{\text{rank}}$ : $\Delta / S / E / C$     | 1.0 / 0.5 / 0.5 / 0.5                   |
| Within $\mathcal{L}_\nu$ : ranking weight                     | 2.0                                     |
| Within $\mathcal{L}_C$ : Stage A consistency weight           | 0.2                                     |
| Within $\mathcal{L}_{\text{dec}}$ : attention / query / usage | 0.7 / 0.1 / 0.2                         |
| Variance floor $\nu$                                          | 0.0025                                  |
| Seed                                                          | 173                                     |

2, and 0 thereafter. This is training-time supervision only: development and test runs always use the upstream joint predictions. Selected joint-coverage features on the global-context path also receive Gaussian noise of standard deviation 0.03 and independent feature dropout at probability 0.10, which leaves untouched both the direct coverage inputs to the novelty heads and the entropy input to the fusion gate.

**Inputs and initialization.** Pair probabilities missing from a supplied ranking record are reconstructed by the input adapter from the available coverage label and score; pair labels and scores also appear in the pair text input. The two facet embeddings entering  $z_G$  have dimension 32 each. The model uses DeBERTa-v3-base with hidden dimension 768, maximum pair-input length 384,  $K = 5$ , and  $M = 8$ ; the residual encoder is initialized from the learned joint model and upstream checkpoints stay fixed. Padded priors have zero support, support values are capped at 0.995 for numerical stability, and an empty evidence set falls back to a target-containing input with all prior support masked out. Fusion uses refinement weights  $(\alpha_D, \alpha_O, \alpha_{MH}) = (0.40, 0.30, 0.30)$  and mixture weights  $(\lambda_D, \lambda_O, \lambda_R) = (0.50, 0.35, 0.15)$  in training; development-set calibration selects  $(0.60, 0.20, 0.20)$  and  $\theta_{MH} = 0.32$  for inference, fixed thereafter. Table 10 lists the remaining settings.

**Evidence-source protocol.** The reference pair pipeline predicts coverage features and ranks priors by the rule in Appendix B.1, while the upstream joint model is trained with bag-level coverage supervision and an auxiliary pair-score regression objective. For the reported checkpoint the evidence order and pair annotations come from the dataset ranking records, so the configuration evaluates novelty prediction with supplied pair annotations and predicted joint coverage. Applying the model to an unseen idea additionally requires computing the pair evidence features.

## B.5 SENSITIVITY OF THE RINOBENCH DECISION CUTS

The EGRD row of Table 3 places its three decision cuts by maximising accuracy on the 1,104 RINoBench train ideas subject to Macro-F1 on that same split staying within a slack  $\epsilon$  of what distribution matching achieves there. Table 11 reports the whole sweep. The cuts are fitted on the train split at every row; only the test columns are shown.

Two things are worth reading off it. Accuracy is not monotone in  $\epsilon$  and moves by five points across the range, so the slack is a real hyperparameter and not a formality. And  $\epsilon = 0.01$ , the value used in the main table, is where Macro-F1 and BalAcc both peak, which means it was not chosen independently of these test numbers. A reader who wants a cut rule with nothing tuned behind it

should take the last row, which maximises accuracy plus Macro-F1 on the train split with no slack parameter at all.

Table 11: Decision-cut sensitivity on RINoBench test, 277 ideas, blockwise minimum of the substantive-delta score over 60 priors read as twelve blocks of five. Distribution matching is the rule used in earlier versions of this table.

| Cut rule                                | Acc           | Ma-F1         | BalAcc        | QWK           | MAE ↓         |
|-----------------------------------------|---------------|---------------|---------------|---------------|---------------|
| Distribution matching                   | 0.3574        | 0.3202        | 0.3214        | 0.2160        | 0.8556        |
| Acc-max, $\epsilon = 0$                 | 0.3718        | 0.3251        | 0.3276        | <b>0.2430</b> | 0.8159        |
| Acc-max, $\epsilon = 0.005$             | 0.3971        | 0.3320        | 0.3372        | 0.2243        | 0.8303        |
| Acc-max, $\epsilon = 0.0075$            | 0.4007        | 0.3350        | 0.3401        | 0.2339        | 0.8159        |
| Acc-max, $\epsilon = 0.01$ (main table) | 0.4152        | <b>0.3407</b> | <b>0.3475</b> | 0.2245        | 0.8087        |
| Acc-max, $\epsilon = 0.015$             | 0.4007        | 0.3074        | 0.3039        | 0.2138        | 0.8014        |
| Acc-max, $\epsilon = 0.02$              | 0.4188        | 0.3172        | 0.3142        | 0.2138        | 0.7798        |
| Acc-max, $\epsilon = 0.03$              | <b>0.4224</b> | 0.3042        | 0.3063        | 0.2048        | <b>0.7798</b> |
| Acc + Ma-F1, no slack parameter         | 0.3971        | 0.3320        | 0.3372        | 0.2243        | 0.8303        |
| same rule at 25 priors                  | 0.4043        | 0.3381        | 0.3430        | 0.2257        | 0.8159        |

## B.6 EVIDENCE BUDGET

The blockwise rule lets the frozen checkpoint, trained with at most five candidate priors per idea, read more evidence than it ever saw in training by running once per block of five and taking the minimum. Raising the budget from 25 priors to 60 moves QWK from 0.2075 to 0.2160 under distribution matching and leaves accuracy unchanged at 0.3574; under the  $\epsilon = 0.01$  rule it moves accuracy from 0.4116 to 0.4152 and QWK from 0.2182 to 0.2245. The gain is real but small, and it saturates: RINoBench ideas carry a median of 23 candidate priors, so beyond roughly 45 the later blocks cover too few ideas to matter. A thirteenth block was not run because only seven test ideas still carry a prior at that depth and the join against the joint-coverage stage returns empty.